15.11 Tensors in Quantum Mechanics

The chapter ends outside mathematics, and the arrangement is the one Chapter 14 used for Minkowski space and for the second derivative test: physics supplies a modeling assumption, the book quotes it and proves nothing about it, and everything after it is linear algebra. The assumption is that a composite quantum system is the tensor product of its parts. Two things follow. One is a phenomenon, **entanglement**, which turns out to be nothing but §02's observation that not every element of $V \otimes W$ is a simple tensor. The other is a theorem, the **Schmidt decomposition**, which turns out to be Chapter 13's singular value decomposition wearing different notation. The second is the real payoff, and it is why this section closes Part IV: it joins the two halves of the book's later chapters at a single point.

15.11.1 What is quoted and what is proved

Quantum mechanics, as a physical theory, is not derived here and cannot be. What we take from it is a short list of modeling conventions, stated once so that the reader can see exactly where the mathematics starts.

(Q1) The states of a quantum system are the **unit vectors** of a complex inner product space V, where two unit vectors differing by a scalar of modulus 1 describe the same state.

(Q2) If a system A has state space V and a system B has state space W, then the composite system "A together with B" has state space $V \otimes W$.

(Q3) If A is in the state v and B is in the state w, then the composite is in the state $v \otimes w$.

None of (Q1)–(Q3) is a theorem, and this book does not argue for any of them. They are quoted exactly as Taylor's theorem was quoted in Chapter 14 §06 — as hypotheses that give the mathematics below something to stand on. Everything proved in this section is a statement about finite-dimensional complex inner product spaces, true whether or not (Q1)–(Q3) describe the world.

Two remarks make the list less arbitrary. First, $\otimes$ rather than $\oplus$ is visible in a dimension count: a spin has a two-dimensional state space, a **qubit**, and two spins together have a four-dimensional one, not a three-dimensional one. Of the two standard constructions, only $\otimes$ produces the observed count, since $\dim(V \otimes W) = \dim V \cdot \dim W$ by Theorem 15.2.5. Second, (Q3) very nearly forces (Q2): the assignment $(v, w) \mapsto$ (joint state) is required to be bilinear, and by Definition 15.2.1 every bilinear map out of $V \times W$ factors uniquely through $V \otimes W$. The universal property is the precise statement that the joint state depends on the parts bilinearly and in no other way.

Genuine quantum mechanics uses infinite-dimensional spaces with a completeness condition. Everything below is finite-dimensional, which is where this book lives, and where qubits live too.

Before states can be unit vectors, $V \otimes W$ needs an inner product. There is only one reasonable candidate, and it exists.

Proposition 15.11.1. The Inner Product on a Tensor Product

Let V and W be finite-dimensional inner product spaces over $F = \mathbb{C}$ (or over $\mathbb{R}$). Then there is **exactly one** inner product on $V \otimes W$ satisfying

$$\langle v \otimes w, v' \otimes w' \rangle = \langle v, v' \rangle \, \langle w, w' \rangle$$

for all $v, v' \in V$ and $w, w' \in W$. Moreover, if $\mathcal{E}$ and $\mathcal{F}$ are orthonormal bases of V and W, then $(e_i \otimes f_j)$ is an orthonormal basis of $V \otimes W$.

Proof. Uniqueness. The simple tensors span $V \otimes W$ (Proposition 15.2.4), and an inner product is additive in the first slot and conjugate-additive in the second, so its values on all pairs of vectors are determined by its values on pairs of simple tensors.

Existence. Fix orthonormal bases $\mathcal{E} = (e_1, \dots, e_m)$ of V and $\mathcal{F} = (f_1, \dots, f_n)$ of W, which exist by Theorem 11.2.2. By Theorem 15.2.5 the mn vectors $e_i \otimes f_j$ form a basis of $V \otimes W$, so the rule

$$\Big\langle \sum_{i,j} c_{ij}\, e_i \otimes f_j,\ \sum_{i,j} d_{ij}\, e_i \otimes f_j \Big\rangle \coloneqq \sum_{i,j} c_{ij}\overline{d_{ij}}$$

is well defined; it is the standard inner product of coordinate vectors, hence satisfies (IP1)–(IP3) of Definition 11.1.1. To check the product formula, write $v = \sum_i a_i e_i$ and $w = \sum_j b_j f_j$. Bilinearity of $\otimes$ gives $v \otimes w = \sum_{i,j} a_i b_j\, e_i \otimes f_j$, and likewise for $v' \otimes w'$, so

$$\langle v \otimes w, v' \otimes w' \rangle = \sum_{i,j} a_i b_j \overline{a'_i b'_j} = \Big(\sum_i a_i \overline{a'_i}\Big)\Big(\sum_j b_j \overline{b'_j}\Big),$$

which is $\langle v, v' \rangle \langle w, w' \rangle$ by Theorem 11.2.1. Finally, the product formula gives $\langle e_i \otimes f_j, e_k \otimes f_l \rangle = \delta_{ik}\delta_{jl}$ for **any** orthonormal $\mathcal{E}$ and $\mathcal{F}$, so the displayed construction did not depend on the pair chosen. This proves the proposition.

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15.11.2 Entanglement is failure of simplicity

The whole phenomenon fits in the language the chapter already has. A state of the composite is a unit vector of $V \otimes W$, and a state of the form $v \otimes w$ is one in which each part has a state of its own. Section 02 proved that not every vector of a tensor product has that form; a composite state that does not describes a system whose parts have no separate states at all.

A state is entangled when it is a vector of $V \otimes W$ that is not a simple tensor.

Definition 15.11.1. Product State, Entangled State

Let V and W be finite-dimensional complex inner product spaces, and let $z \in V \otimes W$ be a unit vector. Then z is a **product state**, also called **separable**, if

$$z = v \otimes w \quad \text{for some } v \in V,\ w \in W,$$

and z is **entangled** otherwise. The same words are used for any nonzero $z \in V \otimes W$, the unit-length condition playing no part in the distinction.

Clause by clause. **Product state** is a synonym for "simple tensor", the term of §02; only the context is new. The definition is relative to the **splitting** $V \otimes W$, not to the vector alone: "entangled" is a statement about a vector *and* a choice of two factors. And v and w need not be unit vectors, though they can be taken so: $\|v \otimes w\| = \|v\|\|w\|$ by Proposition 15.11.1, so rescaling one factor up and the other down costs nothing.

Example. The Bell state

In $\mathbb{C}^2 \otimes \mathbb{C}^2$, write $\mathcal{E} = (e_1, e_2)$ for the standard orthonormal basis of the first factor and $\mathcal{F} = (f_1, f_2)$ for the same basis regarded as sitting in the second factor, the two names being there only to keep the factors apart. Let

$$z_0 \coloneqq \tfrac{1}{\sqrt{2}}(e_1 \otimes f_1 + e_2 \otimes f_2).$$

Show that z_0 is a unit vector, and that it is entangled.

Solution. By Proposition 15.11.1 the four vectors $e_i \otimes f_j$ are orthonormal, so $\|z_0\|^2 = \frac{1}{2}(1+1) = 1$.

For entanglement there is nothing to prove that §02 has not already proved. Once the two bases are recognized as the same one, $e_1 \otimes f_1 + e_2 \otimes f_2$ is exactly the tensor of Example (A tensor that is not simple), and z_0 is a nonzero multiple of it: if $z_0 = v \otimes w$ then $e_1 \otimes f_1 + e_2 \otimes f_2 = (\sqrt{2}\, v) \otimes w$, which Example (A tensor that is not simple) rules out. So z_0 is entangled.

The physical reading is one paragraph long, and the book stops there. Under (Q1)–(Q3), a composite in the state z_0 is a pair of qubits of which two things are true at once: the pair has a completely specified state, and neither member has a state at all, since z_0 is not of the form $v \otimes w$. Nothing about the individual qubits is being suppressed; the postulates simply do not assign them states. That is what is meant by calling the two qubits entangled, and it is why the word attaches to a piece of linear algebra rather than to a measurement. Whether the postulates are true of the world is a question this book does not address.

Warning. **A sum of simple tensors can still be simple.** "Entangled" does not mean "written as a sum of more than one term". In $\mathbb{C}^2 \otimes \mathbb{C}^2$,

$$\tfrac{1}{2}(e_1 \otimes f_1 + e_1 \otimes f_2 + e_2 \otimes f_1 + e_2 \otimes f_2) = \tfrac{e_1+e_2}{\sqrt{2}} \otimes \tfrac{f_1+f_2}{\sqrt{2}},$$

a four-term expression for a product state. Entanglement is a property of the **vector**, not of an expression for it, and deciding it needs an invariant. Supplying one is the job of the rest of this section.

Quick check. Is $z = \frac{1}{\sqrt{3}}(e_1 \otimes f_1 + e_1 \otimes f_2 + e_2 \otimes f_1)$ in $\mathbb{C}^2 \otimes \mathbb{C}^2$ entangled?

Solution. Yes. Suppose $z = v \otimes w$ with $v = a_1 e_1 + a_2 e_2$ and $w = b_1 f_1 + b_2 f_2$. Expanding, $v \otimes w = \sum_{i,j} a_i b_j\, e_i \otimes f_j$, and the $e_i \otimes f_j$ are a basis (Theorem 15.2.5), so comparing coefficients gives

$$a_1 b_1 = a_1 b_2 = a_2 b_1 = \tfrac{1}{\sqrt{3}}, \qquad a_2 b_2 = 0.$$

Then $(a_1 b_1)(a_2 b_2) = 0$ while $(a_1 b_2)(a_2 b_1) = \frac{1}{3}$; but both products equal $a_1 a_2 b_1 b_2$. So no such v, w exist.

15.11.3 The Schmidt decomposition

The Quick check used a 2×2 determinant on an array of coefficients, and that is a hint about where to look. Section 03 already said what the array is: by Theorem 15.3.9 a tensor **is** a linear map, and a linear map between finite-dimensional spaces is a matrix. Chapter 13 knows exactly one thing about an arbitrary matrix, and it knows it completely.

Write out the identification for the case at hand. Fix bases $\mathcal{E} = (e_1, \dots, e_m)$ of V and $\mathcal{F} = (f_1, \dots, f_n)$ of W, and let $(\varphi_1, \dots, \varphi_n)$ be the basis of W^* dual to $\mathcal{F}$ (Theorem 5.1.3). The map

$$V \times W \to \mathcal{L}(W^*, V), \qquad (v, w) \mapsto (\varphi \mapsto \varphi(w)\, v),$$

is bilinear in (v, w), so by Definition 15.2.1 it induces a linear $\Theta\colon V \otimes W \to \mathcal{L}(W^*, V)$. It sends $e_i \otimes f_j$ to the map $\varphi_k \mapsto \delta_{jk} e_i$, whose matrix from $(\varphi_1, \dots, \varphi_n)$ to $\mathcal{E}$ is the matrix unit $\mathbf{E}_{ij}$. The matrix units are a basis of $M_{m\times n}(\mathbb{C})$ and taking the matrix of a map in fixed bases is an isomorphism (Theorem 4.1.4), so Θ carries a basis to a basis and is itself an isomorphism. It is Theorem 15.3.9 with the two factors in the other order and W identified with W^{**} (Theorem 5.4.6), and for coordinate spaces it is §02's matrix model Theorem 15.2.7; the letter is shared for that reason. What matters below is the consequence: the matrix of $\Theta(z)$ is the array of coefficients of z in the basis $(e_i \otimes f_j)$.

Theorem 15.11.2. Schmidt Decomposition

Let V and W be finite-dimensional complex inner product spaces of dimensions m and n, and let $z \in V \otimes W$ be nonzero. Then there are an integer r with $1 \le r \le \min(m, n)$, real numbers

$$\sigma_1 \ge \sigma_2 \ge \cdots \ge \sigma_r > 0,$$

an orthonormal list $(x_1, \dots, x_r)$ in V and an orthonormal list $(y_1, \dots, y_r)$ in W such that

$$z = \sum_{k=1}^{r} \sigma_k\, x_k \otimes y_k. \tag{15.11.1}$$

Moreover r and the numbers $\sigma_1, \dots, \sigma_r$ are determined by z: they do not depend on the lists chosen, nor on any basis. Finally $\|z\|^2 = \sum_k \sigma_k^2$, so z is a unit vector exactly when $\sum_k \sigma_k^2 = 1$.

Idea. The statement is the singular value decomposition of Theorem 13.9.1 with the words changed, so the plan is to change the words back. ① Fix orthonormal bases on both sides; Theorem 15.2.5 turns z into a matrix $\mathbf{C}$ of coefficients, which by the paragraph above is the matrix of the linear map $\Theta(z)$. ② Feed $\mathbf{C}$ to the compact singular value decomposition, Theorem 13.9.3, which over $\mathbb{C}$ still returns **real positive** numbers σ_k and two orthonormal lists of complex column vectors. ③ Read the column vectors back as vectors of V and W; orthonormal columns become orthonormal vectors because coordinates in an orthonormal basis compute inner products. For uniqueness, run ③ backwards: any expression of the shape Eq. 15.11.1 turns into a factorization $\mathbf{C} = \mathbf{X}\mathbf{D}\mathbf{Y}^{\top}$ with $\mathbf{X}, \mathbf{Y}$ having orthonormal columns, and then $\mathbf{C}\mathbf{C}^{*} = \mathbf{X}\mathbf{D}^2\mathbf{X}^{*}$ hands over the σ_k^2 as the nonzero eigenvalues of a matrix built from z alone.

Proof. **Step 1: turn z into a matrix.** Fix orthonormal bases $\mathcal{E} = (e_1, \dots, e_m)$ of V and $\mathcal{F} = (f_1, \dots, f_n)$ of W (Theorem 11.2.2). By Theorem 15.2.5 there are unique scalars $c_{ij} \in \mathbb{C}$ with $z = \sum_{i,j} c_{ij}\, e_i \otimes f_j$. Let $\mathbf{C} \in M_{m\times n}(\mathbb{C})$ be the matrix with entries c_{ij}; it is the matrix of the linear map $\Theta(z)$ described above, and it is nonzero because z is.

Step 2: decompose the matrix. Put $r = \operatorname{rank}\mathbf{C} \geq 1$. By Theorem 13.9.3 there are orthonormal lists $(\mathbf{u}_1, \dots, \mathbf{u}_r)$ in $\mathbb{C}^m$ and $(\mathbf{v}_1, \dots, \mathbf{v}_r)$ in $\mathbb{C}^n$, and the nonzero singular values $\sigma_1 \geq \cdots \geq \sigma_r > 0$ of $\mathbf{C}$ (Definition 13.9.1), with

$$\mathbf{C} = \sum_{k=1}^{r} \sigma_k\, \mathbf{u}_k \mathbf{v}_k^{*}, \qquad \text{that is} \qquad c_{ij} = \sum_{k=1}^{r} \sigma_k (\mathbf{u}_k)_i \overline{(\mathbf{v}_k)_j}.$$

The σ_k are real and positive even though $\mathbf{C}$ is complex, since they are square roots of eigenvalues of the positive semidefinite matrix $\mathbf{C}^{*}\mathbf{C}$. Also $r = \operatorname{rank}\mathbf{C} \leq \min(m, n)$.

Step 3: read the columns back. Set

$$x_k \coloneqq \sum_{i=1}^{m} (\mathbf{u}_k)_i\, e_i \ \in V,$$
$$y_k \coloneqq \sum_{j=1}^{n} \overline{(\mathbf{v}_k)_j}\, f_j \ \in W.$$

Substituting the formula for c_{ij} into $z = \sum_{i,j} c_{ij}\, e_i \otimes f_j$ and interchanging the three finite sums, bilinearity of $\otimes$ gives

$$z = \sum_{k=1}^{r} \sigma_k \Big(\sum_i (\mathbf{u}_k)_i e_i\Big) \otimes \Big(\sum_j \overline{(\mathbf{v}_k)_j} f_j\Big) = \sum_{k=1}^{r} \sigma_k\, x_k \otimes y_k.$$

Since $\mathcal{E}$ is orthonormal, Theorem 11.2.1 gives

$$\langle x_k, x_l \rangle = \sum_i (\mathbf{u}_k)_i \overline{(\mathbf{u}_l)_i} = \langle \mathbf{u}_k, \mathbf{u}_l \rangle = \delta_{kl},$$

and likewise, since $\mathcal{F}$ is orthonormal,

$$\langle y_k, y_l \rangle = \sum_j \overline{(\mathbf{v}_k)_j} (\mathbf{v}_l)_j = \langle \mathbf{v}_l, \mathbf{v}_k \rangle = \delta_{kl}.$$

So both lists are orthonormal, and Eq. 15.11.1 holds.

Step 4: uniqueness. Suppose $z = \sum_{k=1}^{s} \tau_k\, x'_k \otimes y'_k$ is **any** expression of the stated shape, with $\tau_1 \geq \cdots \geq \tau_s > 0$ and both lists orthonormal. Keep the bases $\mathcal{E}, \mathcal{F}$ and write $x'_k = \sum_i \xi_{ik} e_i$ and $y'_k = \sum_j \eta_{jk} f_j$. Let $\mathbf{X} \in M_{m\times s}(\mathbb{C})$ have entries ξ_{ik} and $\mathbf{Y} \in M_{n\times s}(\mathbb{C})$ have entries η_{jk}. Orthonormality of the two lists says, by Theorem 11.2.1 again, that the columns of $\mathbf{X}$ and of $\mathbf{Y}$ are orthonormal, that is

$$\mathbf{X}^{*}\mathbf{X} = \mathbf{I}_s, \qquad \mathbf{Y}^{*}\mathbf{Y} = \mathbf{I}_s.$$

Expanding the expression for z in the basis $(e_i \otimes f_j)$ and comparing with the unique coefficients c_{ij} of Step 1,

$$c_{ij} = \sum_{k=1}^{s} \tau_k\, \xi_{ik} \eta_{jk}, \qquad \text{i.e.} \qquad \mathbf{C} = \mathbf{X}\mathbf{D}\mathbf{Y}^{\top},$$

where $\mathbf{D} = \operatorname{diag}(\tau_1, \dots, \tau_s)$. Taking conjugate transposes and using $\mathbf{Y}^{\top}\overline{\mathbf{Y}} = \overline{\mathbf{Y}^{*}\mathbf{Y}} = \mathbf{I}_s$,

$$\mathbf{C}\mathbf{C}^{*} = \mathbf{X}\mathbf{D}(\mathbf{Y}^{\top}\overline{\mathbf{Y}})\mathbf{D}\mathbf{X}^{*} = \mathbf{X}\mathbf{D}^{2}\mathbf{X}^{*}.$$

Now read off the eigenvalues of $\mathbf{C}\mathbf{C}^{*} \in M_m(\mathbb{C})$. For each k, the k-th column $\mathbf{X}\mathbf{e}_k$ satisfies

$$\mathbf{C}\mathbf{C}^{*}(\mathbf{X}\mathbf{e}_k) = \mathbf{X}\mathbf{D}^{2}\mathbf{X}^{*}\mathbf{X}\mathbf{e}_k = \mathbf{X}\mathbf{D}^{2}\mathbf{e}_k = \tau_k^{2}\,(\mathbf{X}\mathbf{e}_k),$$

and these s columns are orthonormal. If instead $\mathbf{z}' \in \mathbb{C}^m$ is orthogonal to every column of $\mathbf{X}$ then $\mathbf{X}^{*}\mathbf{z}' = \mathbf{0}$, so $\mathbf{C}\mathbf{C}^{*}\mathbf{z}' = \mathbf{0}$. Since $\mathbb{C}^m = \operatorname{col}(\mathbf{X}) \oplus \operatorname{col}(\mathbf{X})^{\perp}$ by Theorem 11.3.2, putting an orthonormal basis of $\operatorname{col}(\mathbf{X})^{\perp}$ beside the columns of $\mathbf{X}$ produces an orthonormal basis of $\mathbb{C}^m$ consisting of eigenvectors of $\mathbf{C}\mathbf{C}^{*}$. Hence the eigenvalue list of $\mathbf{C}\mathbf{C}^{*}$, with multiplicity, is $\tau_1^2, \dots, \tau_s^2$ together with $m - s$ zeros.

Every τ_k^2 is positive, so s is the number of nonzero eigenvalues of $\mathbf{C}\mathbf{C}^{*}$, and $(\tau_1^2, \dots, \tau_s^2)$ is the list of those eigenvalues in decreasing order. Both are determined by $\mathbf{C}$, which is determined by z once $\mathcal{E}$ and $\mathcal{F}$ are fixed. So any two expressions of the shape Eq. 15.11.1 have the same s and the same τ_k; in particular they agree with the r and σ_k of Steps 2 and 3, and since the statement "z admits Eq. 15.11.1 with these r, σ_k" mentions no basis, the numbers cannot depend on $\mathcal{E}$ and $\mathcal{F}$ either.

Step 5: the norm. By Proposition 15.11.1, $\langle x_k \otimes y_k, x_l \otimes y_l\rangle = \langle x_k, x_l\rangle\langle y_k, y_l\rangle = \delta_{kl}$, so the r vectors $x_k \otimes y_k$ are orthonormal in $V \otimes W$. By Theorem 11.1.3, $\|z\|^2 = \sum_k \sigma_k^2$. This proves the theorem.

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Definition 15.11.2. Schmidt Coefficients, Schmidt Rank

Let $z \in V \otimes W$ be nonzero. The numbers $\sigma_1 \geq \cdots \geq \sigma_r > 0$ of Theorem 15.11.2 are the **Schmidt coefficients** of z, and the integer r is the **Schmidt rank** of z. By convention the Schmidt rank of 0 is 0.

Both are well defined precisely because of Step 4, and the proof identifies them: the Schmidt coefficients are the nonzero singular values of the coefficient matrix of z in any pair of orthonormal bases, and the Schmidt rank is the rank of that matrix. That is the invariant the warning above asked for, and it is also §02's: the smallest number of simple tensors summing to z is the rank of $\Theta(z)$, and Theorem 15.11.2 now exhibits a shortest such sum, with its terms orthogonal on both sides at once.

Corollary 15.11.3. Entanglement Is Schmidt Rank at Least Two

A nonzero $z \in V \otimes W$ is a product state if and only if its Schmidt rank is 1. Hence a state is entangled exactly when its Schmidt rank is at least 2.

Proof. ($\Leftarrow$) If $r = 1$ then $z = \sigma_1 x_1 \otimes y_1 = (\sigma_1 x_1) \otimes y_1$, a simple tensor.

($\Rightarrow$) Suppose $z = v \otimes w$. Neither factor is 0, since $z \neq 0$. Put $x_1 = v/\|v\|$ and $y_1 = w/\|w\|$, which is legitimate because $\|v\|, \|w\| > 0$. Then $z = \sigma_1 x_1 \otimes y_1$ with $\sigma_1 = \|v\|\|w\| > 0$, and the one-term lists (x_1), (y_1) are orthonormal. This is an expression of the shape Eq. 15.11.1 with one term, so the uniqueness clause of Theorem 15.11.2 gives $r = 1$.

■

Warning. **The Schmidt coefficients are determined; the vectors are not.** For the Bell state z_0 of Example (The Bell state) the coefficient matrix is $\frac{1}{\sqrt{2}}\mathbf{I}_2$, whose singular values are $\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$. Repeated coefficients leave the lists free, exactly as a repeated eigenvalue of $\mathbf{A}^*\mathbf{A}$ leaves $\mathbf{V}$ free in Theorem 13.9.2: for **every** orthonormal basis (x_1, x_2) of $\mathbb{C}^2$ one has $z_0 = \frac{1}{\sqrt{2}}(x_1 \otimes y_1 + x_2 \otimes y_2)$ for a suitable orthonormal (y_1, y_2). Never speak of *the* Schmidt vectors; speak of *the* Schmidt coefficients.

Example. A Schmidt decomposition in two by three

In $\mathbb{C}^2 \otimes \mathbb{C}^3$, find the Schmidt decomposition, the Schmidt rank, and the Schmidt coefficients of

$$z = \tfrac{1}{2}(e_1 \otimes f_1 + e_1 \otimes f_3 + e_2 \otimes f_2 - e_2 \otimes f_3),$$

where $\mathcal{E}$ and $\mathcal{F}$ are the standard orthonormal bases.

Solution. The coefficient matrix is

$$\mathbf{C} = \tfrac{1}{2}\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \end{pmatrix} \in M_{2\times 3}(\mathbb{R}).$$

This is half the matrix of Example (A singular value decomposition of a 2 by 3 matrix), so by Exercise (B4: Determine whether) (b) its singular values are half of those found there: $\sigma_1 = \frac{\sqrt{3}}{2}$ and $\sigma_2 = \frac{1}{2}$. Both are nonzero, so rank $\mathbf{C} = 2$ and the Schmidt rank is 2; by Corollary 15.11.3, z is entangled. As a check, $\sigma_1^2 + \sigma_2^2 = \frac{3}{4} + \frac{1}{4} = 1$, and indeed $\|z\|^2 = 4 \cdot \frac{1}{4} = 1$.

For the vectors, Theorem 13.9.3 asks for an orthonormal eigenbasis of the 3×3 matrix $\mathbf{C}^*\mathbf{C}$; as in Example (A singular value decomposition of a 2 by 3 matrix) it is cheaper to work with the smaller $\mathbf{C}\mathbf{C}^\top = \frac{1}{4}\left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$, whose orthonormal eigenvectors are the left singular vectors $\mathbf{u}_k$. They are $\frac{1}{\sqrt{2}}(1,-1)$ for the eigenvalue $\frac{3}{4}$ and $\frac{1}{\sqrt{2}}(1,1)$ for $\frac{1}{4}$. So $\mathbf{u}_1 = \frac{1}{\sqrt{2}}(1,-1)$ and $\mathbf{u}_2 = \frac{1}{\sqrt{2}}(1,1)$, and the relation $\mathbf{C}^*\mathbf{u}_i = \sigma_i \mathbf{v}_i$ of Eq. 13.9.3 gives

$$\mathbf{v}_1 = \frac{\mathbf{C}^\top \mathbf{u}_1}{\sigma_1} = \tfrac{1}{\sqrt{6}}(1,-1,2), \qquad \mathbf{v}_2 = \frac{\mathbf{C}^\top \mathbf{u}_2}{\sigma_2} = \tfrac{1}{\sqrt{2}}(1,1,0).$$

All entries are real, so the conjugation in Step 3 of Theorem 15.11.2 does nothing here, and reading the columns back gives

$$\begin{aligned} x_1 &= \tfrac{1}{\sqrt{2}}(e_1 - e_2), \quad y_1 = \tfrac{1}{\sqrt{6}}(f_1 - f_2 + 2f_3), \\ x_2 &= \tfrac{1}{\sqrt{2}}(e_1 + e_2), \quad y_2 = \tfrac{1}{\sqrt{2}}(f_1 + f_2), \end{aligned}$$

so that $z = \frac{\sqrt{3}}{2}\, x_1 \otimes y_1 + \frac{1}{2}\, x_2 \otimes y_2$. Multiplying out recovers z: the two terms are $\frac{1}{4}(e_1 - e_2) \otimes (f_1 - f_2 + 2f_3)$ and $\frac{1}{4}(e_1 + e_2) \otimes (f_1 + f_2)$, whose coefficients at $e_1 \otimes f_2$ and at $e_2 \otimes f_1$ cancel, leaving $\frac{1}{2}$ at $e_1 \otimes f_1$, $e_1 \otimes f_3$, and $e_2 \otimes f_2$ and $-\frac{1}{2}$ at $e_2 \otimes f_3$.

Remark. Truncating Eq. 15.11.1 after k terms is truncating a singular value decomposition, so Theorem 13.11.3 applies verbatim: the closest state of Schmidt rank at most k is the truncated sum, at distance $(\sum_{i>k}\sigma_i^2)^{1/2}$. Exercise (C2: Best approximation by a state of small Schmidt rank) works this out. The remark also marks the limit of the theorem: everything here concerns **two** factors, because a matrix has two indices. For $V \otimes W \otimes U$ a general vector admits no expression as a sum of simple tensors built from three orthonormal families; the book records that without proof.

15.11.4 Looking back at Part IV

Part IV asked one question in two ways. Chapter 14 took a single bilinear form and asked what it is up to a change of basis; Chapter 15 asked what all multilinear maps have in common, and answered by building the space that turns every one of them into a linear map.

The two chapters answer each other. Chapter 14 worked one form at a time and classified: congruence rather than similarity, complete the square, split off a hyperbolic plane, count signs rather than values. Every invariant it produced — the inertia of Theorem 14.5.1, the Witt index of Definition 14.7.4, the discriminant of Definition 14.2.2 — was a number attached to a form that survives a change of basis, and every classification theorem said that a short list of such numbers is all there is. Chapter 15 worked with all forms at once and constructed: define by universal property and then build a model, impose relations by a quotient, define a map out of a quotient upstairs first, and count a basis by counting index tuples. Where Chapter 14 diagonalized a symmetric form by completing the square, Chapter 15 produced the space $\mathrm{Sym}^2 V$ on which that form is a single linear functional; where Chapter 14 proved that alternating is the characteristic-free strengthening of skew, Chapter 15 built $\Lambda^k V$ by killing $v \otimes v$ for exactly that reason; and where Chapter 14 could only say that a Clifford algebra is unique if it exists, Chapter 15 built one out of $T(V)$ and settled its dimension.

This last section adds a third connection, backwards rather than sideways. The Schmidt decomposition is not a new theorem; it is Theorem 13.9.3 with $V \otimes W$ written in place of $M_{m\times n}$. Part III spent a chapter on what a matrix does to two orthonormal bases, one in each of two spaces; Part IV spent a chapter on the space in which a pair of vectors, one from each of two spaces, becomes a single object. Those are the same statement read in two directions, and Theorem 15.3.9 is the sentence that says so. Asked years from now what the tensor product is for, a reader may answer that it is where a bilinear map becomes linear; the shortest true answer is that $V \otimes W$ is where the two sides of a matrix live.

15.11.5 Exercises

15.11.5.1 A. Check your understanding

Exercise. A1

(a) State the three modeling assumptions (Q1)–(Q3) of this section, and say which of them this book proves.

(b) Define **product state** and **entangled state**, and say what data the words depend on besides the vector itself.

(c) State the Schmidt decomposition, with every hypothesis, and say what is unique in it and what is not.

(d) Define the Schmidt rank, and give the criterion for entanglement in terms of it.

(e) True or false: a state written as a sum of three simple tensors has Schmidt rank 3. Justify your answer.

(f) Which theorem of Chapter 13 is the Schmidt decomposition, and which theorem of §03 licenses the translation?

Solution.

(a) (Q1) states are unit vectors of a complex inner product space, up to a scalar of modulus 1; (Q2) the state space of a composite system is the tensor product of the state spaces of the parts; (Q3) if the parts are in states v and w, the composite is in the state $v \otimes w$. The book proves none of them: they are quoted from physics.

(b) A unit vector $z \in V \otimes W$ is a product state if $z = v \otimes w$ for some $v \in V$ and $w \in W$, and entangled otherwise (Definition 15.11.1). The words depend on the **splitting**: the pair of factors V and W, not the vector alone.

(c) Theorem 15.11.2: for V, W finite-dimensional complex inner product spaces and $z \in V \otimes W$ nonzero, there are $r \geq 1$, reals $\sigma_1 \geq \cdots \geq \sigma_r > 0$, and orthonormal lists (x_k) in V and (y_k) in W with $z = \sum_k \sigma_k x_k \otimes y_k$. The number r and the σ_k are unique; the lists are not.

(d) The Schmidt rank is that r (Definition 15.11.2). By Corollary 15.11.3, z is entangled exactly when $r \geq 2$.

(e) False. The Schmidt rank is at most 3, but need not equal 3: the warning after Definition 15.11.1 displays a four-term sum whose Schmidt rank is 1.

(f) It is Theorem 13.9.3, the compact form of the Singular Value Decomposition Theorem 13.9.1. The translation is licensed by Theorem 15.3.9, which identifies a tensor with a linear map and hence, in bases, with a matrix.

15.11.5.2 B. Practice

Exercise. B1: Determine which are entangled

Work in $\mathbb{C}^2 \otimes \mathbb{C}^2$ with the standard orthonormal bases. Determine which of the following unit vectors are entangled. Justify your answer, and for each product state exhibit the two factors.

(a) $\frac{1}{2}(e_1 \otimes f_1 + e_1 \otimes f_2 + e_2 \otimes f_1 + e_2 \otimes f_2)$

(b) $\frac{1}{\sqrt{2}}(e_1 \otimes f_2 - e_2 \otimes f_1)$

(c) $\frac{1}{\sqrt{3}}(e_1 \otimes f_1 + e_1 \otimes f_2 + e_2 \otimes f_2)$

(d) $\frac{1}{\sqrt{10}}(e_1 \otimes f_1 + 2e_1 \otimes f_2 - e_2 \otimes f_1 - 2e_2 \otimes f_2)$

Solution. By Corollary 15.11.3 and the identification of the Schmidt rank with the rank of the coefficient matrix, it suffices to compute $\operatorname{rank} \mathbf{C}$ in each case, and for a 2×2 matrix that is decided by the determinant.

(a) $\mathbf{C} = \frac{1}{2}\left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, with $\det \mathbf{C} = 0$ and $\mathbf{C} \neq \mathbf{O}$, so $\operatorname{rank} \mathbf{C} = 1$: a product state. Indeed it equals $\frac{1}{\sqrt{2}}(e_1 + e_2) \otimes \frac{1}{\sqrt{2}}(f_1 + f_2)$.

(b) $\mathbf{C} = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, with $\det \mathbf{C} = \frac{1}{2} \neq 0$, so $\operatorname{rank} \mathbf{C} = 2$: entangled.

(c) $\mathbf{C} = \frac{1}{\sqrt{3}}\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, with $\det \mathbf{C} = \frac{1}{3} \neq 0$, so $\operatorname{rank} \mathbf{C} = 2$: entangled.

(d) $\mathbf{C} = \frac{1}{\sqrt{10}}\left(\begin{smallmatrix} 1 & 2 \\ -1 & -2 \end{smallmatrix}\right)$, with $\det \mathbf{C} = 0$ and $\mathbf{C} \neq \mathbf{O}$, so $\operatorname{rank} \mathbf{C} = 1$: a product state, equal to $\frac{1}{\sqrt{2}}(e_1 - e_2) \otimes \frac{1}{\sqrt{5}}(f_1 + 2f_2)$.

Exercise. B2: A Schmidt decomposition

In $\mathbb{C}^2 \otimes \mathbb{C}^2$, let

$$z = \tfrac{1}{\sqrt{10}}(e_1 \otimes f_1 + 2e_1 \otimes f_2 + 2e_2 \otimes f_1 + e_2 \otimes f_2).$$

Find the Schmidt coefficients, the Schmidt rank, and a Schmidt decomposition of z, and verify that $\sum_k \sigma_k^2 = 1$.

Solution. The coefficient matrix is $\mathbf{C} = \frac{1}{\sqrt{10}}\left(\begin{smallmatrix} 1 & 2 \\ 2 & 1 \end{smallmatrix}\right)$, which is real symmetric. Then

$$\mathbf{C}\mathbf{C}^{\top} = \tfrac{1}{10}\begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix},$$

with characteristic polynomial $x^2 - x + \frac{9}{100}$ and eigenvalues $\frac{9}{10}$ and $\frac{1}{10}$. So the Schmidt coefficients are $\sigma_1 = \frac{3}{\sqrt{10}}$ and $\sigma_2 = \frac{1}{\sqrt{10}}$, the Schmidt rank is 2, and $\sigma_1^2 + \sigma_2^2 = \frac{9}{10} + \frac{1}{10} = 1$.

Eigenvectors of $\mathbf{C}\mathbf{C}^{\top}$ are $\mathbf{u}_1 = \frac{1}{\sqrt{2}}(1, 1)$ for $\frac{9}{10}$ and $\mathbf{u}_2 = \frac{1}{\sqrt{2}}(1, -1)$ for $\frac{1}{10}$. Then

$$\mathbf{v}_1 = \frac{\mathbf{C}^{\top}\mathbf{u}_1}{\sigma_1} = \tfrac{1}{\sqrt{2}}(1, 1), \qquad \mathbf{v}_2 = \frac{\mathbf{C}^{\top}\mathbf{u}_2}{\sigma_2} = \tfrac{1}{\sqrt{2}}(-1, 1).$$

All entries are real, so reading the columns back as in Step 3 of Theorem 15.11.2,

$$x_1 = \tfrac{e_1+e_2}{\sqrt{2}}, \quad y_1 = \tfrac{f_1+f_2}{\sqrt{2}}, \qquad x_2 = \tfrac{e_1-e_2}{\sqrt{2}}, \quad y_2 = \tfrac{-f_1+f_2}{\sqrt{2}},$$

and $z = \frac{3}{\sqrt{10}}\, x_1 \otimes y_1 + \frac{1}{\sqrt{10}}\, x_2 \otimes y_2$. Expanding the two terms gives $\frac{1}{2\sqrt{10}}(3 - 1)e_1 \otimes f_1$ and similarly for the other three coefficients, recovering z.

Exercise. B3: How large can the Schmidt rank be?

Let V and W be complex inner product spaces with $\dim V = m$ and $\dim W = n$.

(a) Prove that every nonzero $z \in V \otimes W$ has Schmidt rank at most $\min(m, n)$.

(b) Give, for each r with $1 \leq r \leq \min(m, n)$, a unit vector of Schmidt rank exactly r.

Solution.

(a) Fix orthonormal bases and let $\mathbf{C} \in M_{m\times n}(\mathbb{C})$ be the coefficient matrix of z. By Theorem 15.11.2 the Schmidt rank is $\operatorname{rank} \mathbf{C}$, and the rank of an $m \times n$ matrix is at most $\min(m, n)$ because $\operatorname{rank} \mathbf{C} = \dim \operatorname{col}(\mathbf{C}) \leq m$ and $\operatorname{rank} \mathbf{C} = \dim \operatorname{row}(\mathbf{C}) \leq n$.

(b) Take $z_r = \frac{1}{\sqrt{r}} \sum_{k=1}^{r} e_k \otimes f_k$. The lists $(e_1, \dots, e_r)$ and $(f_1, \dots, f_r)$ are orthonormal and every coefficient $1/\sqrt{r}$ is positive, so this is already an expression of the shape Eq. 15.11.1; by the uniqueness clause the Schmidt rank is r. It is a unit vector since $r \cdot (1/\sqrt{r})^2 = 1$.

15.11.5.3 C. Going deeper

Exercise. C1: Acting on one part at a time

Let $S \in \mathcal{L}(V)$ and $T \in \mathcal{L}(W)$ be unitary, and let $z \in V \otimes W$ be nonzero.

(a) Prove that $(S \otimes T)(z)$ has the same Schmidt coefficients, and hence the same Schmidt rank, as z.

(b) Hence deduce that no operator of the form $S \otimes T$ with S, T unitary carries the Bell state z_0 of Example (The Bell state) to a product state.

Hint for (a): apply $S \otimes T$ to a Schmidt decomposition.

Solution.

(a) Write $z = \sum_{k=1}^{r} \sigma_k \, x_k \otimes y_k$ as in Theorem 15.11.2. By Definition 15.3.1, $(S \otimes T)(x_k \otimes y_k) = Sx_k \otimes Ty_k$, so by linearity

$$(S \otimes T)(z) = \sum_{k=1}^{r} \sigma_k \, (Sx_k) \otimes (Ty_k).$$

A unitary operator preserves inner products (Theorem 11.7.3), so $\langle Sx_k, Sx_l \rangle = \langle x_k, x_l \rangle = \delta_{kl}$ and likewise for T. Both lists are therefore orthonormal, the σ_k are still positive and decreasing, and the display is an expression of the shape Eq. 15.11.1 for $(S \otimes T)(z)$. By the uniqueness clause of Theorem 15.11.2, its Schmidt coefficients are $\sigma_1, \dots, \sigma_r$ and its Schmidt rank is r.

(b) The Schmidt rank of z_0 is 2, as computed in the warning after Corollary 15.11.3. By (a), $(S \otimes T)(z_0)$ also has Schmidt rank 2, and by Corollary 15.11.3 a product state has Schmidt rank 1. So $(S \otimes T)(z_0)$ is entangled.

Exercise. C2: Best approximation by a state of small Schmidt rank

Let $z \in V \otimes W$ be nonzero with Schmidt decomposition $z = \sum_{k=1}^{r} \sigma_k \, x_k \otimes y_k$, and let $0 \le k \le r$. Put $z_{(k)} = \sum_{i=1}^{k} \sigma_i \, x_i \otimes y_i$.

(a) Fix orthonormal bases of V and W and let $\mathbf{C}$ be the coefficient matrix of z. Prove that $\|z\| = \|\mathbf{C}\|_F$, and that the Schmidt rank of a vector equals the rank of its coefficient matrix.

(b) Hence deduce that for every $z' \in V \otimes W$ of Schmidt rank at most k,

$$\|z - z'\| \;\ge\; \Big(\sum_{i>k} \sigma_i^2 \Big)^{1/2},$$

with equality for $z' = z_{(k)}$.

Solution.

(a) By Proposition 15.11.1 the basis $(e_i \otimes f_j)$ of $V \otimes W$ is orthonormal, so $\|z\|^2 = \sum_{i,j} |c_{ij}|^2$ by Theorem 11.9.2, and that sum is $\|\mathbf{C}\|_F^2$ by the definition of the Frobenius norm. For the second claim, Step 2 of Theorem 15.11.2 produced the Schmidt coefficients as the nonzero singular values of $\mathbf{C}$, and Theorem 13.9.1 says there are rank $\mathbf{C}$ of them.

(b) The map $z \mapsto \mathbf{C}$ is the coordinate isomorphism attached to the basis $(e_i \otimes f_j)$ (Corollary 2.5.5), with the mn coordinates arranged in a rectangle, so it is linear and bijective; and by (a) it carries $\|\cdot\|$ to $\|\cdot\|_F$ and Schmidt rank to matrix rank. Let $\mathbf{B}$ be the coefficient matrix of z', so $\operatorname{rank} \mathbf{B} \le k$. By Theorem 13.11.3,

$$\|z - z'\| = \|\mathbf{C} - \mathbf{B}\|_F \;\ge\; \Big(\sum_{i>k} \sigma_i^2\Big)^{1/2},$$

since the nonzero σ_i are the singular values of $\mathbf{C}$. Running Step 3 of Theorem 15.11.2 backwards turns $z_{(k)}$ into the rank-k truncation $\mathbf{C}_k$ (Definition 13.11.1) of a compact singular value decomposition of $\mathbf{C}$ (Theorem 13.9.3), and Theorem 13.11.3 attains equality there.

Exercise. C3: Does the field matter?

Let V and W be **real** inner product spaces of dimensions m and n, with orthonormal bases $\mathcal{E}$ and $\mathcal{F}$, and let $z \in V \otimes W$ have coefficient matrix $\mathbf{C} \in M_{m\times n}(\mathbb{R})$. Let $z_{\mathbb{C}} \in \mathbb{C}^m \otimes \mathbb{C}^n$ be the vector whose coefficient matrix in the standard orthonormal bases is the same $\mathbf{C}$.

(a) Prove that Theorem 15.11.2 holds verbatim over $\mathbb{R}$, with orthonormal lists of real vectors.

(b) Prove that z and $z_{\mathbb{C}}$ have the same Schmidt coefficients and the same Schmidt rank.

(c) Is the same true of the Schmidt **vectors**? Justify your answer.

Solution.

(a) Every step of the proof runs over $F = \mathbb{R}$. Theorem 11.2.2, Theorem 15.2.5, Theorem 13.9.3, Theorem 11.2.1, and Theorem 11.3.2 are all stated for $F = \mathbb{R}$ or $\mathbb{C}$, and over $\mathbb{R}$ the conjugations in Step 3 and Step 4 are the identity, so $y_k = \sum_j (\mathbf{v}_k)_j f_j$ and $\mathbf{C}\mathbf{C}^{\top} = \mathbf{X}\mathbf{D}^2\mathbf{X}^{\top}$.

(b) By Step 4 of Theorem 15.11.2, read over each field in turn, the Schmidt coefficients of z and of $z_{\mathbb{C}}$ are both times the positive square roots of the nonzero eigenvalues of $\mathbf{C}\mathbf{C}^{\top}$, and both Schmidt ranks equal $\operatorname{rank} \mathbf{C}$. This is one and the same real symmetric matrix with one and the same characteristic polynomial, so the eigenvalues agree; and $\operatorname{rank} \mathbf{C}$ is unchanged because row reduction over $\mathbb{R}$ is row reduction over $\mathbb{C}$.

(c) No. The warning after Corollary 15.11.3 says why: a repeated Schmidt coefficient leaves the vectors free, and over $\mathbb{C}$ there is more freedom to use. For z_0 the orthonormal pair $x_1 = \frac{1}{\sqrt{2}}(e_1 + ie_2)$, $x_2 = \frac{1}{\sqrt{2}}(e_1 - ie_2)$ occurs in a complex Schmidt decomposition —

with $y_1 = \frac{1}{\sqrt{2}}(f_1 - if_2)$ and $y_2 = \frac{1}{\sqrt{2}}(f_1 + if_2)$ — and has no real counterpart. What is invariant is the coefficient list, not the vectors.